

HIGHLY INTEGRATED HDTV PROCESSOR WITH MULTI-CODEC

DESCRIPTION

EMMA3TL2 is a HDTV processor supporting worldwide digital/analog broadcasting formats. It integrates in a single chip the TV set core, which includes the CPU, AV signal processing, panel signal processing, peripheral control functions, and more. It is ideal for Full HDTV architectures.

FEATURES

- Multi-CPU system (Main CPU, Sub CPU, Extended CPU)
- DDR2 memory interface (x32, DDR2-800 technology up to 3.2 GB/s)
- Stream processor supporting W/W scrambling signal
Encryption/decryption: DVB, AES, Multi2, DES and 3DES functions
Supported streams: MPEG2-TS and MPEG1-System, DVB, ARIB, ATSC
- Network DRM Support for IP Streaming (Windows DRM, etc.)
- Audio DSP supporting multi-audio format systems and Audio Description (MPEG BC, Dolby™ Digital, MPEG2 AAC, MPEG2 HE-AAC, Dolby™ Digital Plus, DRA,)
- Virtual surround
- Multi format video decoder (MPEG1, MPEG2, MPEG-4 AVC/H.264, VC-1, AVS, DivX-HD, MPEG4-ASP)
Bandwidth saving decode function, DC-IL support
- High-performance graphics engine, 3-plane 32bpp OSD (main video: 2 planes, sub-video: 1 plane)
- De-interlacer supporting Full HD
- 7 x 5 tap high-performance video scaler
- Super resolution function integrated
- 15-bit signal processing picture quality adjustment controller (Picture Quality Controller) supporting xvYCC
LTI/CTI, 2D aperture, sharpness, APL adaptive black extension/white peak correction, contrast, hue correction, Liner – γ , MATRIX conversion, V-T correction, LCD overdrive, FRC, etc.
Dual Link LVDS output supporting Full HD (for main video, 1920H x 1080 V x 60 Hz, 30-bit GBR)
- Downconvert analog video output (PAL/SECAM/NTSC: YC or CVBS Output)
- 4Ch high speed video ADCs with analog video switch supporting up to 1080P and WUXGA
- 3D-Comb analog video decoder with SCART support, SECAM flicker free filter
- Picture in picture (PC&CVBS) support
- HDMI version 1.3a receiver
- Stereo Audio ADC with 6 in - 1out analog switch and 4 stereo DACs
- Various external interfaces
(Ethernet, USB2.0, Smart Card, Memory Stick, 3-UART, 3-IIC, CSI Host, GIO, GPIO, etc.)
- OneNAND I/F
- CI Buffer integrated
- Various on-chip PLLs, clock input: 48.000 MHz for System
- Power supply voltage: 3.3 V, 1.8 V (for external interface, analog core), 1.0 V (for PLL, HDMI, internal logic core)

ORDERING INFORMATION

Part Number	Package
MC-10157	632-pin plastic BGA (1.0mm pitch) (31x31)

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BLOCK DIAGRAM

